

Homework #2

CSED501

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Due: **October 28, 2025**Name: Aeden Jameson

Instructions: Complete the questions below and upload a PDF containing your written answers to Gradescope.

- Keep your answers within the provided space.
- Answers can be handwritten, or typed and pasted into this PDF.
- Show your work; this makes giving partial credit easier!
- True/False questions can be answered with **True/False**, but explanations can be added if you feel they are needed.
- Numeric answers can be given in fraction or decimal form; if decimal, round to 2 places

Assigned reading (links can also be found in Canvas Week 4 Resources):

- Friedler, Sorelle A., Carlos Scheidegger, and Suresh Venkatasubramanian. "The (Im) possibility of Fairness: Different Value Systems Require Different Mechanisms For Fair Decision Making What does it mean to be fair?." (2021).

Problem 1 6 pts: *True/False Questions*

- (a) Using PCA for dimensionality reduction provides a subset of the original features and is generally interpretable.
- (b) Logistic regression minimizes the sum of squared errors (residuals).
- (c) Decision trees are less likely to overfit than random forests.
- (d) AdaBoost is the best choice for classifying data with extreme outliers.
- (e) When using PCA for dimensionality reduction on a training dataset of size n by D , where n is the number of training samples and D is the number of initial features, you pick the k principal components from the n possible principal components created by PCA.
- (f) Increasing the regularization parameter C in an SVM makes the model more tolerant of misclassified points.

Answer:

- a) False. PCA creates features that are combinations of existing features and not a strict subset which results in those features being less interpretable.
- b) False. Logistic regression optimizes the log-likelihood function.
- c) False. Generally decision trees are high variance models which means they will be sensitive to slight changes in data. Thus the model has overfit the data.
- d) False. AdaBoost is sensitive to outliers.
- e) False. There are not n components but D . n designates the number of rows.
- f) False. Increasing C results in more weight on misclassification.

Problem 2 5 pts: Consider the following confusion matrix:

	Negative(actual)	Positive(actual)
Negative(predicted)	125	80
Positive(predicted)	75	120

- (a) Is the dataset balanced?
- (b) What are the numbers of TP, TN, FP, and FN?
- (c) What is the accuracy?
- (d) What are the precision and recall?
- (e) What is the F1 score?

Answer:

a) Assuming the total $TP + TN + FP + FN = n$ where n is the total number of points in the full training set, then this dataset is balanced because,

- * $TP + FN \sim TN + FP$
- * $120 + 80 \sim 125 + 75$
- * $200 = 200$

Both values of the response variable occur equally.

Problem 3 **9 pts**: You're supervising a team, and one of your employees tells you about a plan to train a random forest classifier in parallel on a CPU with 8 cores. This employee details the plan as follows:

- (a) Put a copy of the whole training dataset on each core.
- (b) Train a decision tree on that dataset with all of the features.
- (c) Use the majority vote of the trees to choose a class.

For each step of the plan, write 1-2 sentences explaining either why the suggested step is correct, or why the suggested step is wrong/suboptimal along with what should be done instead.

Answer:

- a) This step is from a practical standpoint incorrect. First, each cpu core alone typically doesn't have it's own memory. There are exceptions to this, but nonetheless the size of the training set could very well be too large as well.
- b) Incorrect. The general guidance is to use a subset of features.
- c) Correct. The majority vote algorithm is a standard algorithm for selecting the class label.

Problem 4 **10 pts**: The authors of “The (Im) possibility of Fairness” describe two possible worldviews: What You See is What You Get (WYSIWYG) and We’re All Equal (WAE).

- (a) In your own words, describe the difference between the two worldviews.
- (b) Using an example not found in the article or class, propose a situation in which following WYSIWYG and WAE would lead to different outcomes. Describe the potential different outcomes and discuss their practical consequences.

Answer:

- a) The core difference between the WYSIWYG and WAE worldviews is in their ability to ensure individual fairness or non-discrimination with respect to a fairness goal. WYSIWYG assumes the observed state of affairs and decisions made from that state are what ought to be, known as the construct space. In contrast WAE assumes all groups are equal in the construct space and that the observed differences are largely a product of circumstances not under individual control.
- b) Interviewing of software engineers has steered heavily towards what are called LeetCode style questions. The questions focused heavily on algorithms and are timed. The WYSIWYG assumption is that the timing and format will select the best individuals for the job. In my experience WYSIWYG selects for test takers and unproductively competitive individuals in the work place. WAE view would focus more questions sampled from the work needed by the role. I've found it leads to finding people that are a better fit for the work.

Problem 5 12 pts:

You are the Chief Research Officer of a personal loan company that commits to ethical and transparent decision-making. The company's stated goal is to use explainable models that are aware of and mitigate systematic bias in loan approvals. Your proprietary data goes 60 years back and includes hundreds of features, such as income history, gender, race, credit score, zip code, criminal record, prior loan defaults, employment history, and dozens of additional behavioral and financial indicators which are not essential for answering the questions below. You want to predict whether an applicant will be able to pay back the loan they are currently requesting based on these attributes.

Your team proposes the following plan to maximize profit while staying true to the company's stated objective:

- (a) Train very deep trees to maximize accuracy, but use all available 60 years of historical data in an attempt to improve generalization and prevent overfitting.
- (b) Remove obvious sensitive attributes such as gender and race, but keep all other financial and behavioral features.
- (c) Present decision paths(all feature splits from root to leaf) in a user-friendly manner to applicants to justify approvals or denials.
- (d) Use overall accuracy as the primary metric guiding model selection.

For each step of the team's proposal, write 1–2 sentences arguing either:

- That the step is appropriate and why
- That the step is inappropriate, a suggestion for what to do instead and why

Answer:

a) Inappropriate. That sixty years of data will contain historical biases. Additionally, deep tree overfit and so when applied to new applicants (unseen data) there's a general risk of the model not performing well. Instead more shallow trees should be trained on more recent data and incorporate a non-discrimination goal for the model.

b) Inappropriate. Dropping those fields isn't likely to achieve non-discrimination because proxy variables most likely exist. Instead keep the fields for the purposes having a model the company can audit for systemic bias.

c) Inappropriate. Doing this would be a branding and/or credibility risk. A small group of loud people that are a genuine risk can attack the brand or because the model isn't perfect it will say no on some close calls. Instead give straight forward response like is often done today. e.g. Bad credit, Insufficient income.

d) Inappropriate. As we saw in the homework accuracy can hide class imbalance. Instead use techniques to deal with class imbalance: re-weighting, re-sampling or SMOTE (seems popular, but I have no experience with it)

Problem 6 16 pts:

Answer each of the following questions with 1-2 sentences or a mathematical formula where appropriate.

- (a) You train a logistic regression classifier on inputs (x_1, x_2) , but first expand them with all degree-2 polynomial features. For this two-dimensional input $x = (x_1, x_2)$ and the degree-2 polynomial expansion, explicitly write the function $P(y = 1|x)$
- (b) Can we, in general (so not necessarily for polynomial features), compare $|w_i|$ across features to judge which features are the most important for classification? Explain why or why not.
- (c) Suppose you train a regularized logistic regression model with L_2 penalty. During training via gradient descent, does the log-likelihood of the data necessarily increase at every iteration? Why/why not?
- (d) For softmax logistic regression with K classes we the parameter vectors $w_1, \dots, w_K$ and the model

$$P(y = k|x) = \frac{e^{w_k^\top x}}{\sum_{j=1}^K e^{w_j^\top x}}.$$

Suppose for a fixed input x the model predicts class $k^* \in \{1, \dots, K\}$. That is, $k^* = \arg \max_k P(y = k|x)$. What can you say about the values $(w_{k^*} - w_k)^\top x$ for any class k not equal to k^* ?

Answer:

$$a) \quad P(y=1|x) = \frac{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1^2 + \beta_4 x_1 x_2 + \beta_5 x_2^2}{e^{-\epsilon}}$$

where $x = (x_1, x_2)$

- b) No I don't think which features are the most important can easily be determined for the simple reason that features can have different units.

c) Not sure

d)

Problem 7 **10 pts**: In 1–2 paragraphs, discuss a problem that you’re interested in solving (this could be related to a work problem, a potential personal project, or just something that interests you) that could be solved with classification. What kind of data might you start by collecting? What classification algorithm(s) would you try, and why?

Answer:

Problem: Will a pull request be approved or need changes?

Description: Predict whether or not pull request will be approved with minor comments or be blocked with needs changes?

Data: The data that would be helpful in predicting the time would include: author, lines of code, # reviewers, #files changed, # previous approvals, # previous needs changes, date, the repository, problematic coding constructs

Model: I don't have a good intuition about which algorithm would perform best so I would end up trying them all.

Problem 8 14 pts: You're working on a problem to classify incoming emails into three inboxes: general, important, and spam. You decide to use logistic regression, using only one feature, the number of words in all caps included in the email.

Take an example feature vector $x = [1, 2]$, where $x_0 = 1$ is the intercept term and $x_1 = 2$ is the number of words in all caps. We'll provide you with learned model parameters for both binary (One-vs-Rest) and multinomial logistic regression:

$$w_0 = [-1, 1], \quad w_1 = [-2, 2], \quad w_2 = [1, -0.5].$$

You can use any calculator that you would like to compute $e^{-\hat{y}}$ and $e^{\hat{y}}$; round each $e^{\hat{y}}$ to 2 decimal places before computing probabilities.

- (a) For binary logistic regression using one-vs-rest, you train a separate binary classifier for each class k , estimating

$$P(y = k | x) = \frac{1}{1 + e^{-w_k^T x}}.$$

First, write the probability from each binary classifier in the form $P(y = k | x) = \frac{1}{1 + e^{-\hat{y}}}$, with your computed values for $w_k^T x$ plugged in for $\hat{y}$. Then find the probability as a decimal value rounded to two places from each binary classifier. Finally, identify the predicted class of x .

- (b) For multinomial logistic regression using softmax, you directly estimate the probabilities

$$P(y = k | x) = \frac{e^{w_k^T x}}{\sum_{j=0}^2 e^{w_j^T x}}.$$

First, write the probability from each class in the form $P(y = k | x) = \frac{e^{\hat{y}}}{\sum_{j=0}^2 e^{\hat{y}}}$, with your computed values for $w_k^T x$ plugged in for $\hat{y}$. Then find the probability as a decimal value rounded to two places for each class. Finally, identify the predicted class of x .

Answer:

a) First compute values from linear model,

$$w_0^T x = 1 \cdot 1 + 1 \cdot 2 = 3 \quad (\text{general})$$

$$w_1^T x = 2 \cdot 1 + 2 \cdot 2 = 6 \quad (\text{important})$$

$$w_2^T x = 1 \cdot 1 + (-0.5) \cdot 2 = 0 \quad (\text{spam})$$

The probabilities are

$$\text{general} = \frac{1}{1+e^{-3}} = .05, \text{ important} = \frac{1}{1+e^{-6}} \approx .9$$

$$\text{spam} = \frac{1}{1+e^{-2}} = .88$$

Prediction is "important"

$$b) \text{ general} = \frac{20.09}{20.01 + 403.43 + 7.39} = .05$$

$$\text{important} = \frac{403.43}{430} = .94$$

$$\text{spam} = \frac{7.39}{430} = .02$$

Prediction = important.

Problem 9 **15 pts**: Decision trees can be used for regression as well as classification. In this problem, you will walk through how the inner workings of the decision tree would need to change to output a continuous value (like the price of a house) instead of a class. We'll refer to decision trees for classification tasks as *classification trees*, and decision trees for regression tasks as *regression trees*.

- (a) Classification trees choose splits based on metrics such as Gini impurity. What kind of metric might a regression tree choose splits on instead (there are several correct answers here)? Would it minimize or maximize that metric?
- (b) A "good" split for a classification tree is one where a high proportion of data points are from a single class. How would a "good" split be defined for a regression tree (without referencing specific quality metrics)?
- (c) In a classification tree, the prediction at a leaf node is often the majority class. In a regression tree, what might the predicted value at a leaf node be, and why?

Answer:

a) A regression tree can minimize the mean squared error.

b) A good split for a regression tree would

- * Not have overlapping, or too much, values between children.
- * No node is impacted by outliers

c) In a regression tree the value at the leaf node would be the mean of the response stored in the leaf because mean minimizes the error.

Problem 10 **3 pts**: How many hours do you estimate you spent on this assignment? (graded on completion of (a), (b), and (c))

(a) Coding 5 hours

(b) Written 5 - 6 hours - There's a lot struggling with tooling though.

(c) Reading 10 hours

Answer:

Problem 11 **Gratitude pts**: Any feedback you would like to share (optional)?

Answer: Can we work on the written part in a Jupyter notebook? Overleaf and the tools to work with PDF's are very tedious and time consuming.